DV2626/DV2599 Machine Learning

Lecture 10: Weakly Supervised Learning and Multi-instance Learning

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Outline

- Introduction to weakly and multi-instance learning
  - Weakly supervised learning
  - MIL setup
  - Standard MIL assumption
  - Use Diversity Density to solve MI problems
  - Application of MIL
- Unsupervised multi-instance learning
- Multi-instance clustering: BAMIC algorithm
- BARTMIP multi-instance prediction

Strong vs Weak Supervision

- Types of weak supervision:
  - **incomplete supervision**, i.e., only a subset of training data are given with labels whereas the other data remain unlabeled
  - **inexact supervision**, i.e., only coarse-grained labels are given
  - **inaccurate supervision**, i.e., the given labels are not always ground-truth

- **Weakly supervised learning** is an attempt to construct predictive models by learning with weak supervision

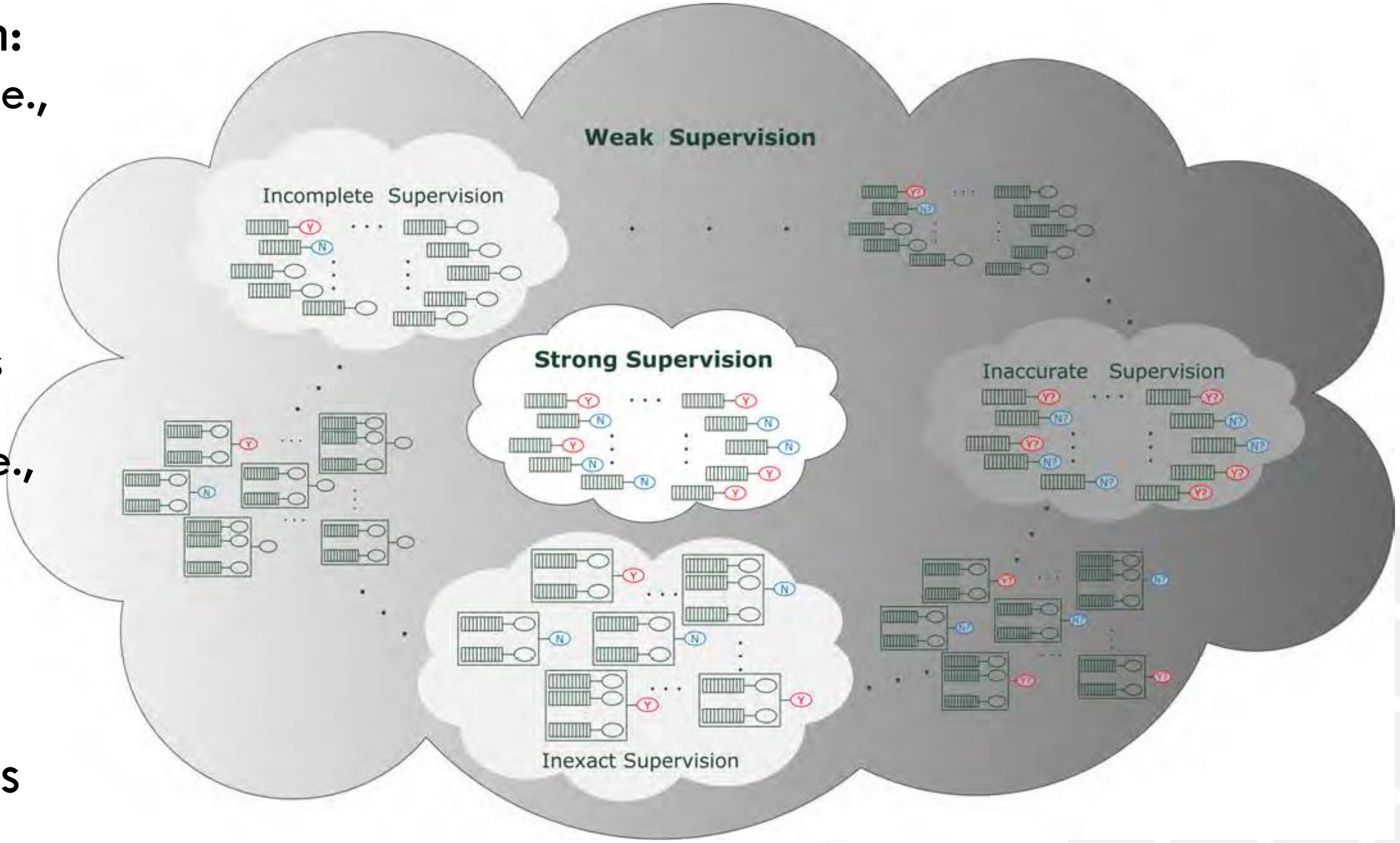
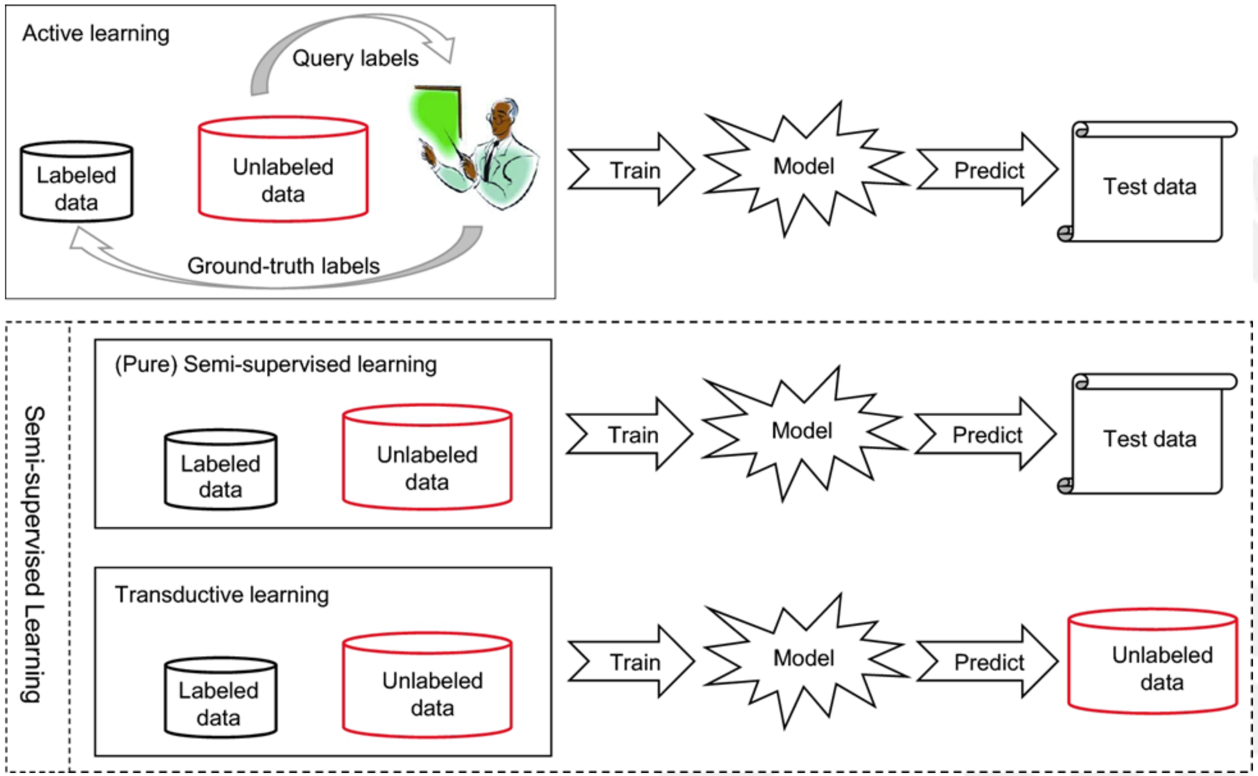


Image from: Zhi-Hua Zhou, A brief introduction to weakly supervised learning, *National Science Review*, Volume 5, Issue 1, January 2018, Pages 44–53.

Incomplete supervision

- **Active learning** assumes that the ground-truth labels of unlabeled instances can be queried from an oracle
- **Semi-supervised learning** attempts to exploit unlabeled data without querying human experts
- **Transductive learning** is semi-supervised learning that attemps to predict unlabeled data by the trained model



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Active learning

- The goal of **active learning** is to minimize the number of queries to reduce labeling cost

- Two main selection criteria:
  - **Informativeness** measures how well an unlabeled instance helps reduce the uncertainty of a statistical model
    - Uncertainty sampling
    - Query-by-committee
  - **Representativeness** measures how well an instance helps represent the structure of input patterns
    - Clustering

Semi-supervised learning

- Clustering assumption
  - instances falling into the same cluster have the same class
- Manifold assumption
  - data lie on a manifold, and thus, nearby instances have similar predictions

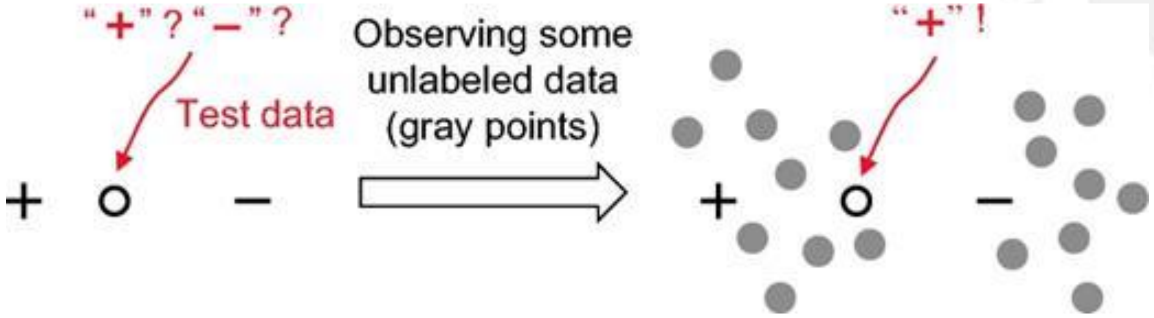


Image from: Zhi-Hua Zhou, A brief introduction to weakly supervised learning, *National Science Review*, Volume 5, Issue 1, January 2018, Pages 44–53.

Inaccurate supervision

- Label noise filtering techniques
- Techniques for label noise identification and relabeling
- Rely on a relative neighborhood graph

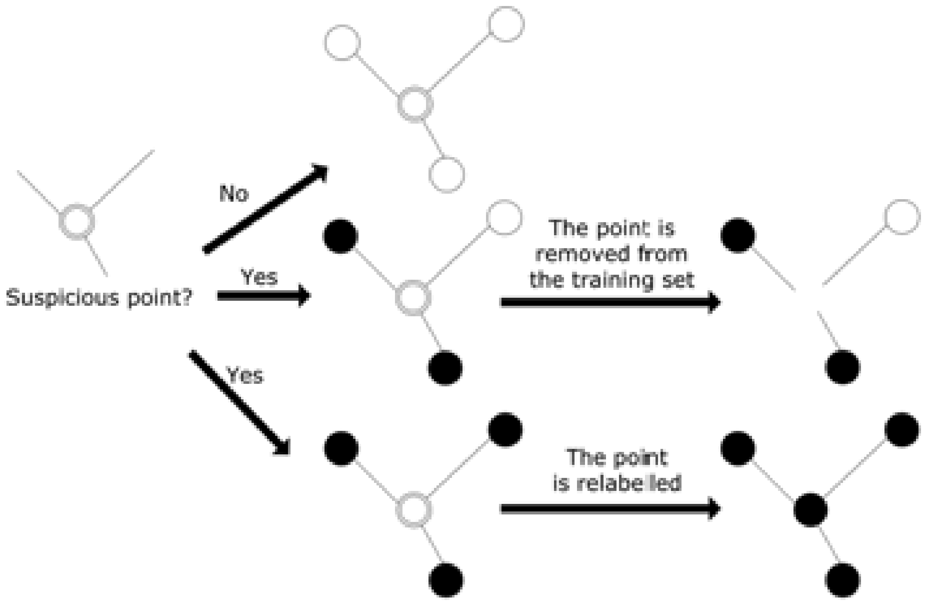


Image from: Zhi-Hua Zhou, A brief introduction to weakly supervised learning, *National Science Review*, Volume 5, Issue 1, January 2018, Pages 44–53.

Inexact supervision: *multi-instance learning*

- **Multi-instance learning** (MIL) is a form of weakly supervised learning
- Training instances are arranged in **sets**, called **bags**
- A label is provided for entire bags **but not for instances**

Multi-instance learning motivation

- Some problems are **naturally formulated** as MIL
  - drug activity prediction
  - scene classification
  - web mining
  - …

- It is gaining momentum in the pattern recognition community because:
  - It deals with **weakly annotated data**
  - This reduces the annotation cost, since collecting labeled data is often expensive
  - Algorithms can now learn from a greater quantity of training data

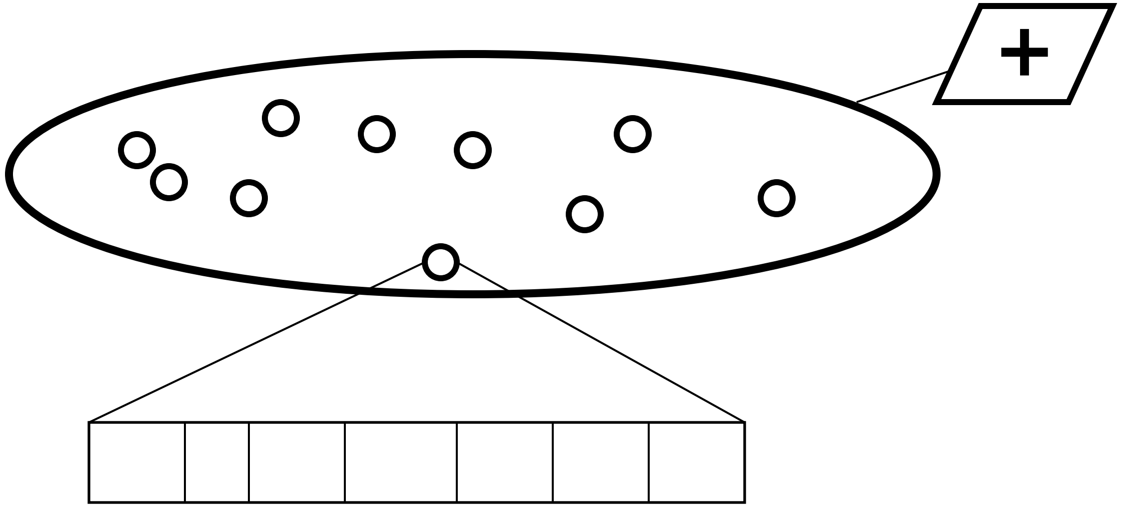
Multi-instance learning setup

Given: A set of labeled data objects (training set) and unlabeled data objects (test set)

Bag: Each data object is a bag and is associated with a label ('+' or '-')

Instance: A **bag** is described by a set of instances.
Each instance is described by a vector of features.
Every instance also has a (hidden/unknown) label.

Problem: Predict the class of an unlabelled bag.

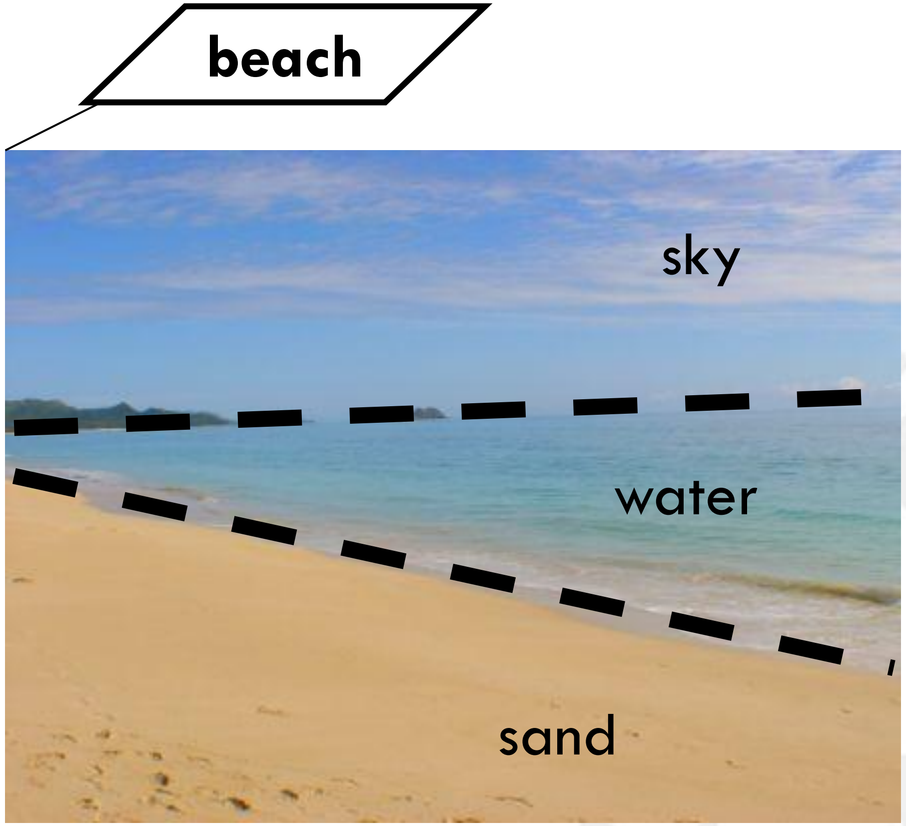


Multi-instance learning example: scene classification

Bag -> An image

Instance -> A 'region' in the image

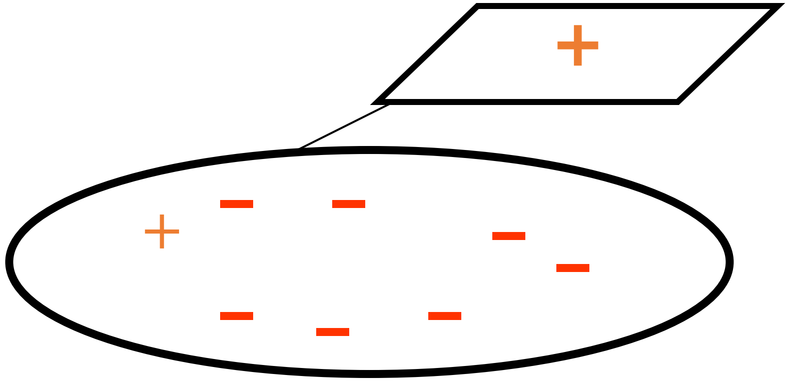
Label -> {'beach', 'not beach'}

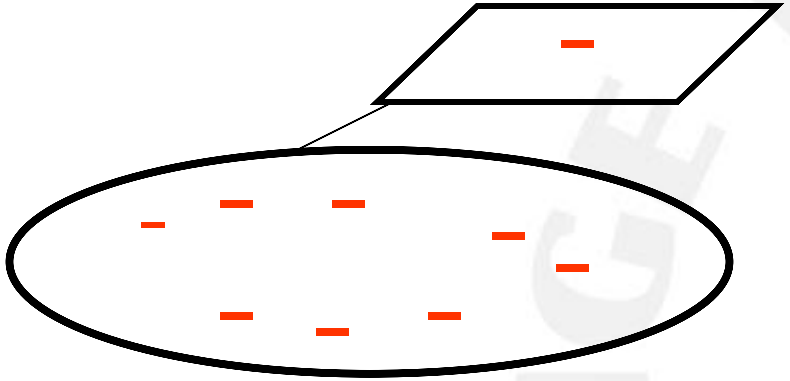


http://www.adrhi.com/Waimanalo-Beach.jpg

Standard MIL assumption

- **Negative** bags do not contain positive instances
- Positive bags may contain negative and positive instances
- **Positive** bags contain at least one positive instance





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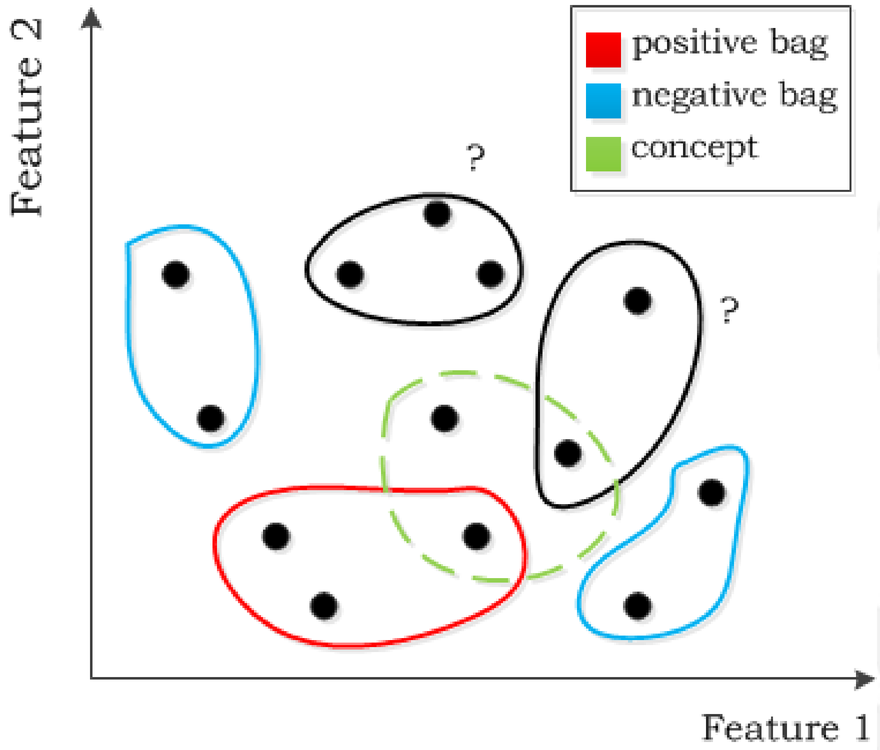
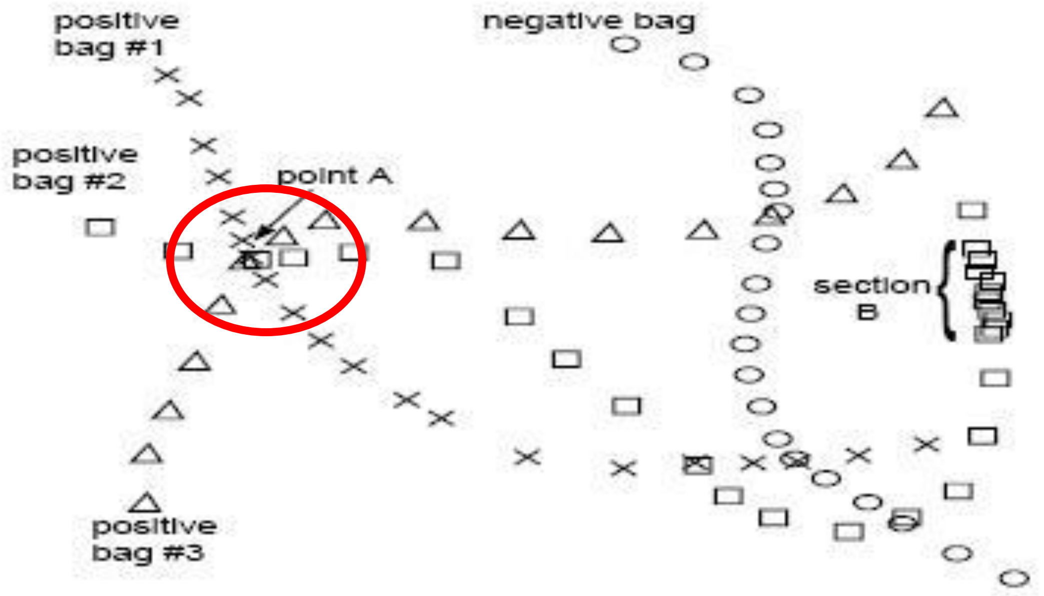


Use Diversity Density to solve MI problems

Diverse Density (DD) is a measure of the intersection of the positive bags minus the union of the negative bags.

DD of an instance measures how close an instance is to instances of different positive bags while also being far away from all instances of negative bags.

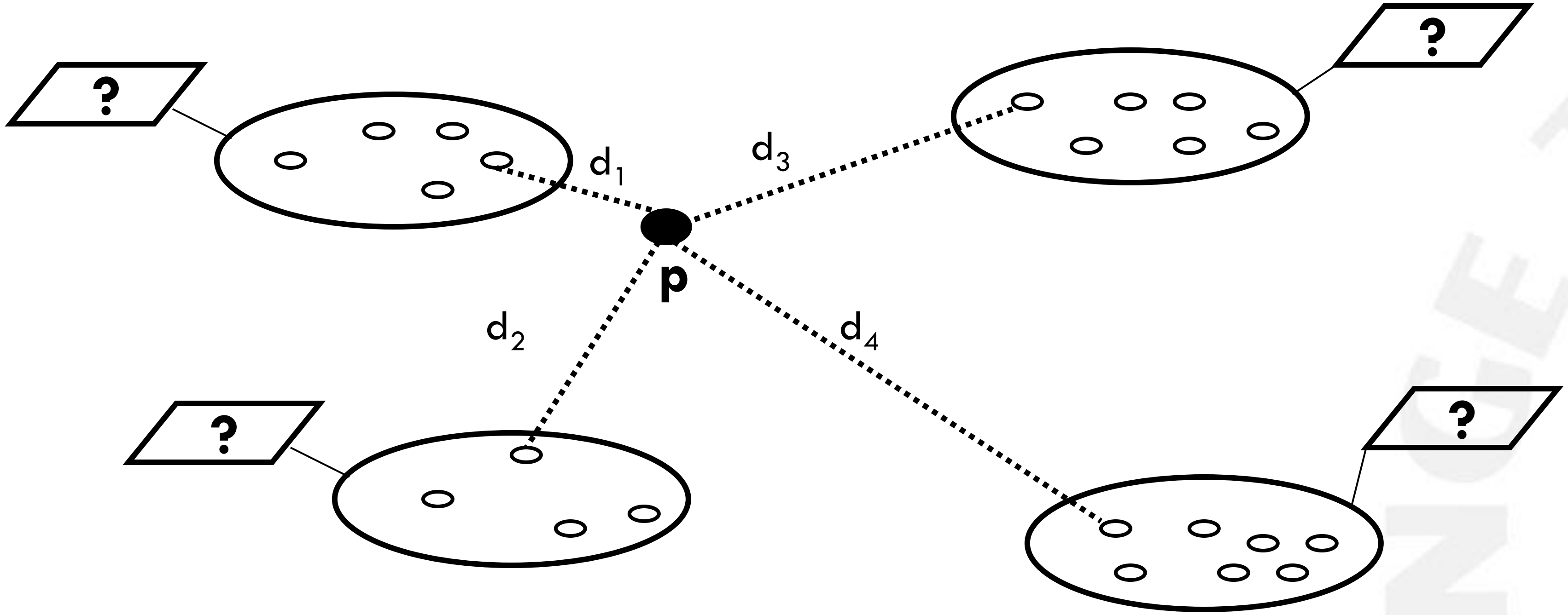
The instance with the maximum DD value is the most positive instance.



O. Maron and T. Lozano-Pérez. A framework for multiple-instance learning. In Advances in Neural Information Processing Systems 10, pages 570-576. Cambridge, MA: MIT Press, 1998

Use Diversity Density to solve MI problems

- Find the point (named *p*) in instance space with the maximum DD value.
- Use the max DD value to compute a distance threshold *t*.
- A test bag *b* with instances x_i is classified as positive if $\min_i(\text{distance}(x_i, p)) < t$



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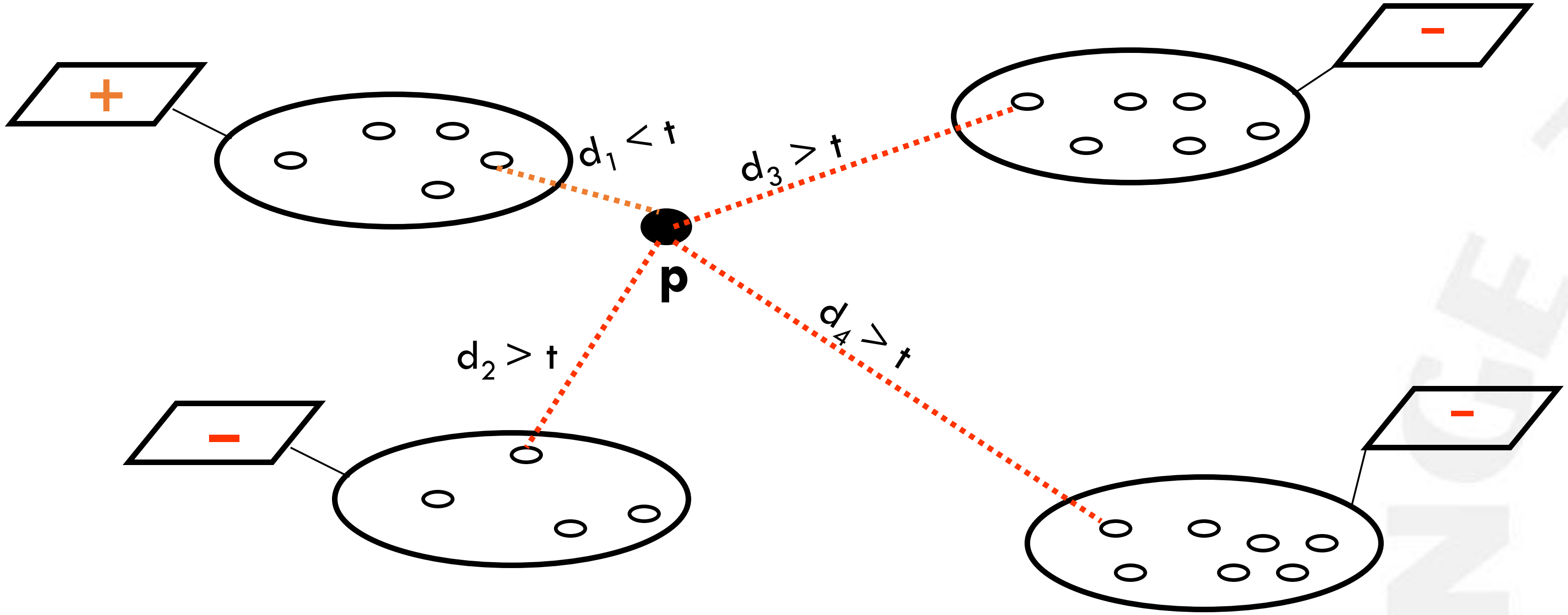
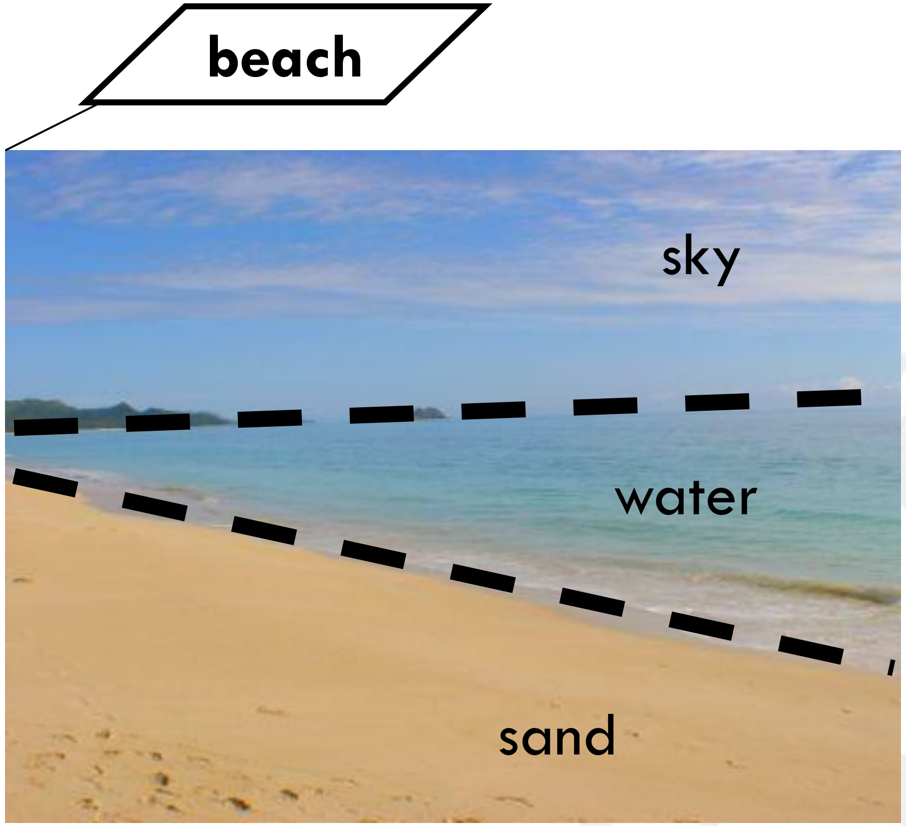


Multi-instance learning example: scene classification



Bag -> An image

Instance -> A 'region' in the image

Label -> {'beach', 'not beach'}

http://www.adrhi.com/Waimanalo-Beach.jpg

A 'beach scene' could contain "water" or "sand" or "sky".

Example of relaxed MIL assumptions

- Both **sand** and **water** segments are positive instances for beach pictures.
- However, picture of beach must contain **both segments** of sand and water. Otherwise, they can be pictures of desert or sea.

Image from: J. Amores, "Multiple instance classification: Review, taxonomy and comparative study," *Artif. Intell.*, vol. 201, pp. 81–105, Aug. 2013.

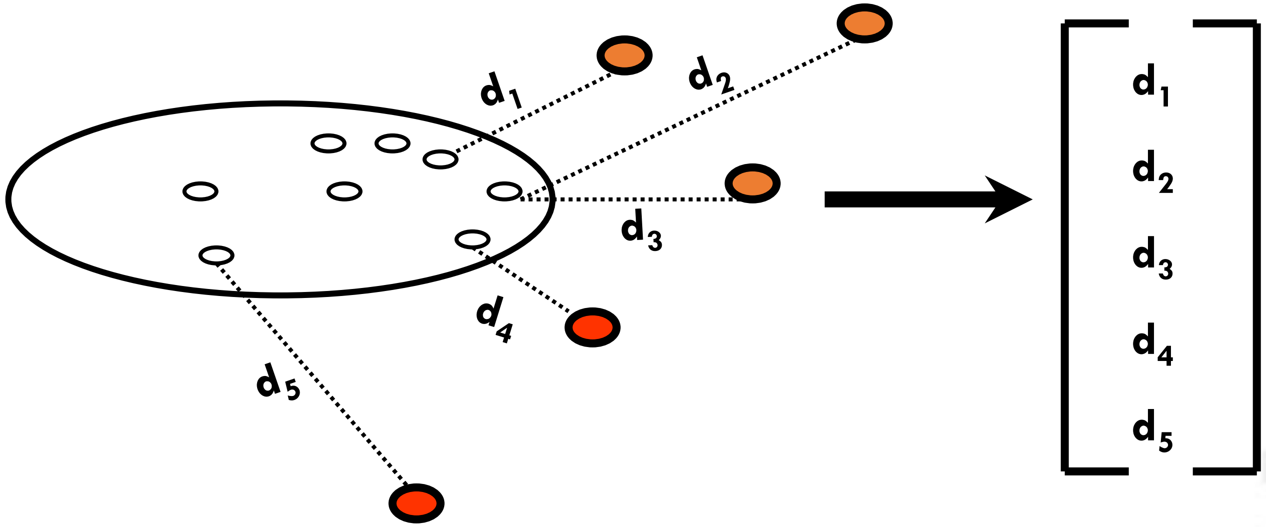
Relaxed MIL assumptions

- In many applications, the standard MIL assumption is too restrictive.
- MIL can alternatively be formulated as:
  - A bag is positive when it contains a sufficient number of positive instances.
  - A bag is positive when it contains a certain combination of positive instances.
  - Positive and negative bags differ by their instance distributions.

J. Fouldsand E. Frank, “A Review of Multi-Instance Learning Assumptions,” *Knowl. Eng. Rev.*, vol. 25, no. 1, pp. 1–25, Mar. 2010.

Use DD-SVM to solve MI problems

- The DD-based approach uses one instance point (the one with the globally maximal DD value) as a prototype for all positive instances.
- DD-SVM approach represents the positive instance space by a set of **instance prototypes** (locally max DD values).
- A bag is represented by a vector of minimum distances to each of the prototypes.

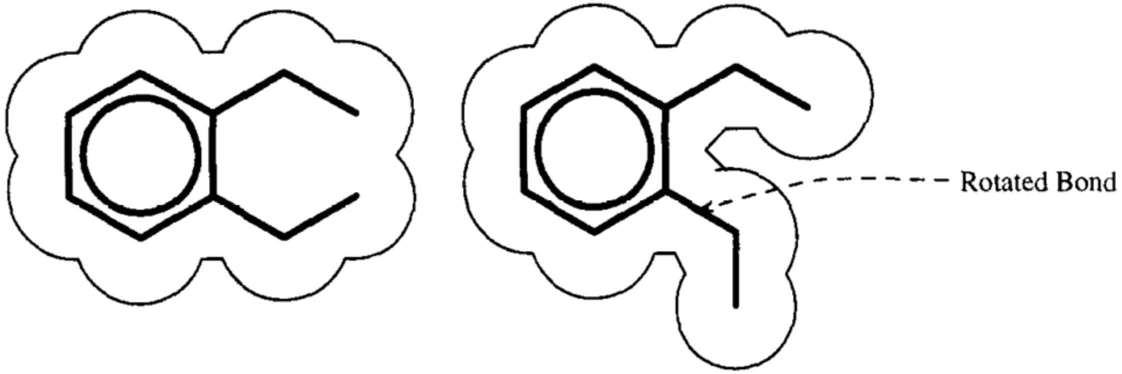


Train a SVM in the "bag-feature" space.

Y. Chen and J. Wang, image categorization by learning and reasoning with regions, The Journal of Machine Learning Research (2004) 5, 913-939

Application of MIL: Drug Activity Prediction

- **Objective:** Predict if a molecule produces a given effect.
- **Bag:** Collection of all conformations of the same molecule.
- **Instance:** Conformation of a molecule.
- **Justification:** Conformations are not observable individually.



T. G. Dietterich, R. H. Lathrop, and T. Lozano-Pérez, "Solving the Multiple Instance Problem with Axis-parallel Rectangles," *Artificial Intelligence* 1997.

Application of MIL: Computer Aided Diagnosis (from images)

- **Objective:** Predict if a subject is diseased or healthy.
- **Bag:** Collection of segments or patches extracted from a medical image.
- **Instances:** Image segments or patches.
- **Justification:** A large quantity of images can be used to train. Only a diagnosis is required per image. Expert local annotation is no longer required.

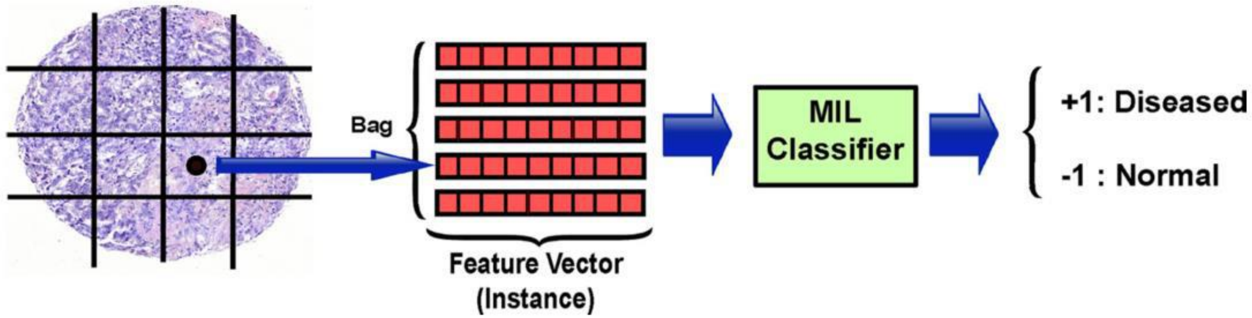


Image from: M. Kandemirand F. A. Hamprecht, "Computer-aided diagnosis from weak supervision: a benchmarking study.," *Comput. Med. Imaging Graph.*, vol. 42, pp. 44–50, Jun. 2015.

Application of MIL: Content Base Image Retrieval

- **Objective:** Classify images based on their subject
- **Bag:** Collection of segments or patches extracted from an image
- **Instances:** Image segments or patches
- **Justification:** Images can represent composite objects or concepts

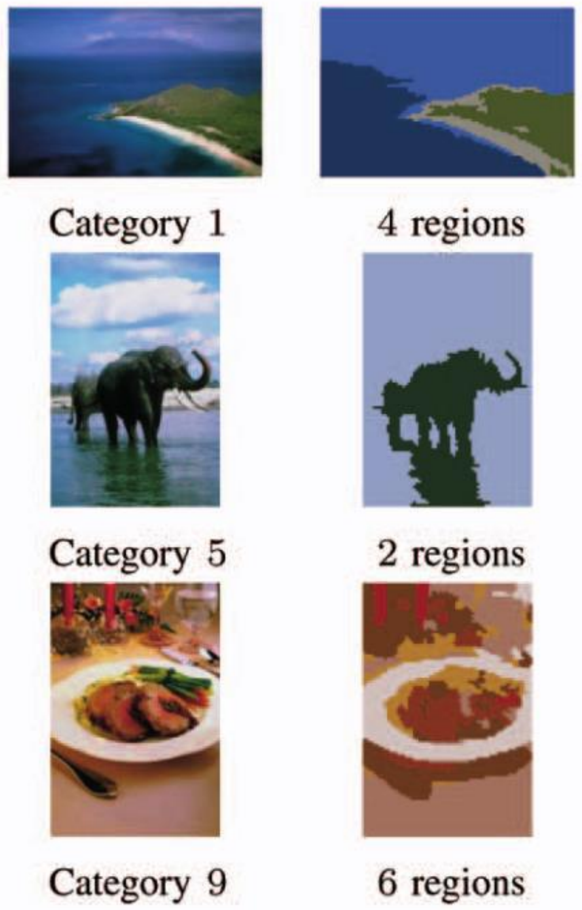


Image from: Y. Chen, J. Bi, and J. Z. Wang, “MILES: Multiple-Instance Learning via Embedded Instance Selection,” *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 28, no. 12, pp. 1931–1947, 2006.

Application of MIL: Sentiment Analysis in Text

- **Objective:** Predict if a text/sentence expresses positive or negative sentiment.
- **Bags:** Texts/paragraphs
- **Instances:** Sentences
- **Justification:** Large quantity of text can be harvested from the web. A sentiment is usually given to a complete text while it may contain positive and negative sentences.

> Paul Bettany did a great role as the tortured father whose favorite little girl dies tragically of disease.
> For that, he deserves all the credit.
> However, the movie was mostly about exactly that, keeping the adventures of Darwin as he gathered data for his theories as incomplete stories told to children and skipping completely the disputes regarding his ideas.
> Two things bothered me terribly: the soundtrack, with its whiny sound, practically shoving sadness down the throat of the viewer, and the movie trailer, showing some beautiful sceneries, the theological musings of him and his wife and the enthusiasm of his best friends as they prepare for a battle against blind faith, thus misrepresenting the movie completely.
> To put it bluntly, if one were to remove the scenes of the movie trailer from the movie, the result would be a non descript family drama about a little child dying and the hardships of her parents as a result.
> Clearly, not what I expected from a movie about Darwin, albeit the movie was beautifully interpreted.

Image from: D. Kotzias, M. Denil, P. Blunsom, and N. de Freitas, "Deep Multi-Instance Transfer Learning," *CoRR*, vol. abs/1411.3, 2014.

Unsupervised MIL

- **Unsupervised MIL** is a descriptive task where the learning process is carried out without information about the labels of bags.

- The necessity of unsupervised MIL
  - In some cases, it is hard or costly to obtain labels for the bags
  - Unsupervised learning could help find the inherent structure of a data set

- It is more complex than traditional single-instance unsupervised learning due to the inherent ambiguity in MIL.

F. Herrera et al., *Multiple Instance Learning*, Springer International Publishing AG 2016 DOI 10.1007/978-3-319-47759-6_7.

MI Clustering

- **Multi-instance (MI) clustering** splits up a set of unlabeled bags into a number of homogeneous subgroups on the basis of a similarity measure.

- **MI Clustering** has its own characteristics and the similarity measures used in single-instance clustering may not be appropriate.

MI Clustering example

- In **drug activity prediction**, only one or a few of the conformations describing a molecule would be responsible for its qualification.
- Only few of the regions describing an image is useful in the **object identification task.**

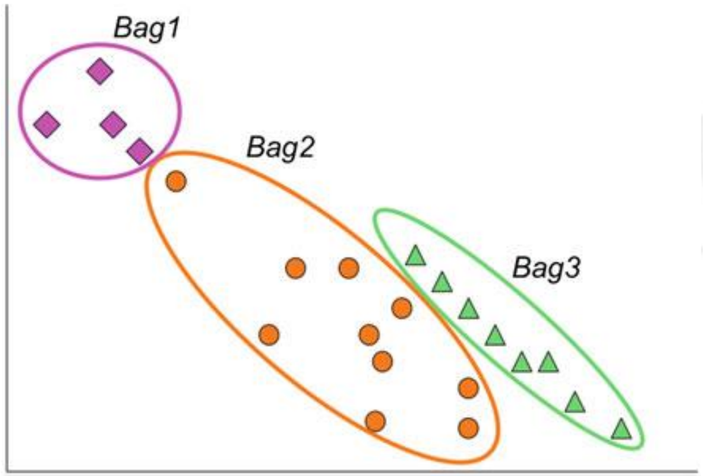


Image from: F. Herrera et al., *Multiple Instance Learning*, Springer International Publishing AG 2016.

The natural ambiguity of MI-Clustering is about which portion of the image (bag) contains the clustering concept and which portion may be irrelevant.

Some facts about MI Clustering

- The bags should not be regarded as simple collections of independent instances.
- The characteristics and relationships of the instances within bags should be carefully investigated.
- It is critical to identify the desired objects/concepts in each bag, such that the unsupervised distance definitions can reveal the true distances between bags.

BAg-level Multi-Instance Clustering (BAMIC)

- **BAMIC** extends the popular *k*-medoids algorithm to complete multi-instance clustering tasks.
- It partitions the unlabeled training bags into *k* disjoint groups of bags.
- It uses a bagwise distance called **Hausdorff distance.**

Zhang, M., Zhou, Z. Multi-instance clustering with applications to multi-instance prediction. *Appl Intell* **31**, 47–68 (2009). https://doi.org/10.1007/s10489-007-0111-x

Average, Max and Min Hausdorff distances

- Given two bags of instances $A = \{a_1, \ldots, a_m\}$ and $B = \{b_1, \ldots, b_n\}$, the *average Hausdorff distance* is defined as

$$\mathrm{aveH}(A, B) = \frac{\sum_{a \in A} \min_{b \in B} \|a - b\| + \sum_{b \in B} \min_{a \in A} \|b - a\|}{|A| + |B|}$$

- *Maximal and minimal Hausdorff distances* are defined as

$$\mathrm{maxH}(A, B) = \max\left\{\max_{a \in A} \min_{b \in B} ||a - b||, \max_{b \in B} \min_{a \in A} ||b - a||\right\}$$

$$\mathrm{minH}(A, B) = \min_{a \in A, b \in B} \|a - b\|$$

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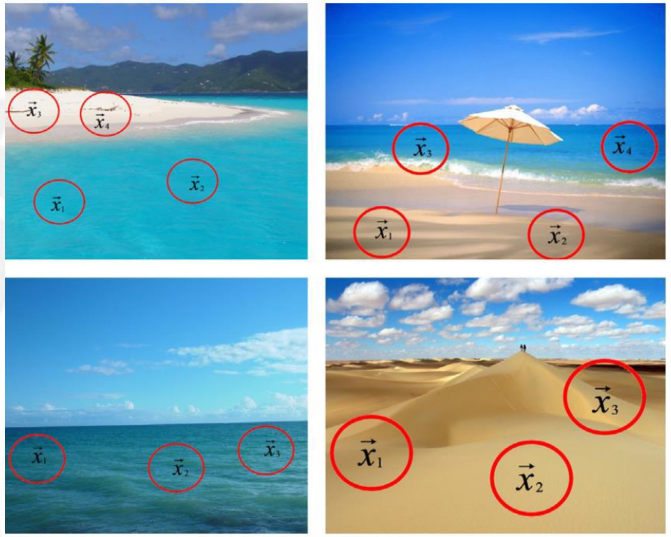

Multi-Instance Prediction (1)

- Two strategies to deal with the representation difficulty encountered in multi-instance prediction
  - Modify supervised learning algorithms to meet the representation of feature vector sets
  - Transforming the representation to meet the requirement of common supervised learning algorithms

Multi-Instance Prediction (2)

- Two groups of multi-instance prediction algorithms based on representation transformation
  - bags are transformed into corresponding feature vectors by structuring the input space at the level of *instances*
  - representation transformation is performed by structuring the input space at the level of *bags*

BAg-level Representation Transformation for Multi-Instance Prediction (BARTMIP)

- Training bags are firstly clustered into *k* disjoint *groups of bags* using BAMIC
- All the bags are transformed into the *k*-dimensional feature vectors
  - Each bag is re-represented by a *k*-dimensional feature vector whose *i*-th feature corresponds to the distance between the bag and the medoid of the *i*-th group

- Common supervised learners could be used to train on the generated feature vectors to distinguish the bags
- When an unseen bag is given for prediction, it is firstly transformed into the corresponding feature vector through measuring its distance to different group medoids

Questions?